

Research Report

A Data-Driven Framework for Understanding Social and Structural Drivers of Adolescent Vaccination Uptake to Inform Targeted Outreach

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Abstract

Background: The uptake of recommended vaccinations among adolescents is affected by various socioecological characteristics. By integrating the Social Ecological Model (SEM) with a predictive model, we developed a machine learning based data-driven framework to identify adolescents at risk of not being up to date (UTD) on recommended vaccinations (Tdap, HepB, MMR, MenACWY, varicella, and HPV series). The framework further generated risk stratified profiles to support targeted outreach strategies.

Methods: A retrospective analysis was conducted using 2023 National Immunization Survey Teen data (N=17,021 adolescents aged 13-17 years). Multiple predictive models were evaluated using a 70/30 training and validation split. The optimal model was selected through cross-validated hyperparameter tuning. Significant predictors were interpreted within the SEM framework, and adolescents were classified into risk tiers with associated socioecological profiles.

Results: The optimal model was a LASSO regression with hyperparameter tuning, achieving AUC of 0.68 on the validation set (F1 score = 0.73). SEM aligned positive predictors of adolescent vaccination UTD status were related to organization and structural domains, including healthcare access, preventive care utilization, and provider recommendation. Geographic context (predominantly northern states) and individual/family characteristics (adolescent health status, parental income, family support, race/ethnicity) were also significant. Predicted probabilities were grouped into five tiers, with observed UTD rates declining from 85.5% (lowest risk) to 27.3% (highest risk), demonstrating clear risk separation.

Conclusions: This data-driven framework emphasized the important role of provider engagement and equitable healthcare access along with geographic and socio-economic factors, reinforcing the need for targeted, multilevel efforts to improve adolescent vaccination completion.

Background

Vaccines are a cornerstone of public health, keeping people and communities safe from preventable diseases. Among adolescents (aged 13-17 years), being up to date with vaccines such as HPV (Human Papillomavirus Vaccine), Tdap (Tetanus, Diphtheria, and Acellular Pertussis), HepB (Hepatitis B), MMR (Measles, Mumps, and Rubella Vaccine), MenACWY (Meningococcal Conjugate Vaccine, serogroups A, C, W, and Y), varicella, and seasonal influenza vaccines are crucial for long term health (ACIP 2025). In the United States (U.S.), the recommended series of vaccines during this age range include complete HPV vaccine series if not finished earlier, continue receiving the annual influenza vaccine, and follow current COVID-19 vaccination recommendations. A booster dose of meningococcal ACWY is recommended at age 16 to ensure continued protection against meningococcal disease. Some adolescents in this age group may also be eligible for meningococcal B (MenB) vaccination (based on shared clinical decision-making) (ACIP 2025, CDC 2025). Effective January 2025, federal health officials recommended focusing on MMR, polio, HPV, and chickenpox, while shifting other vaccines (Hep A/B, Flu, COVID-19) to high-risk population or shared clinical decision-making (HHS 2026). However, despite expert recommendations, vaccination rates in adolescents are suboptimal and have shown mixed trends for different vaccines, with recent data comparing 2024 to 2023 showing an increase in Tdap vaccination rates while vaccination rates for MenACWY and HPV stagnated (Pingali 2025). More importantly, vaccinations took a major hit during COVID-19 pandemic and the rates are now starting to normalize. It is critical that they continue to move in that direction.

Previous studies have identified several social and behavioral drivers of vaccine uptake among adolescents including their age, parents' perceived barriers, mis-information about vaccines, education level, income, and access to healthcare (Li 2020, Vardavas 2023, Sacre 2023, Sonawane 2024). The disparities in vaccination behavior are impacted by social context, healthcare accessibility, and exposure to health information. These factors can also predict the future intentions of taking vaccines, thus understanding the dynamics of these determinants is crucial for successful outreach efforts to improve vaccination rates among adolescents.

Recent evidence suggests that vaccine uptake rates are driven by multilevel ecology of beliefs, trust, and access barriers meaning broad, one-size-fits-all campaigns often miss the actual reasons people delay or refuse vaccines. Enyinnaya et al. (2024) developed the Social

Ecology of Health Beliefs and Misinformation Framework by combining the Health Belief Model (HBM) with the Socio-Ecological Model (SEM) using interviews and focus groups with physicians, public health professionals, and parents. This study reported that misinformation can negatively impact vaccine uptake through multiple layers simultaneously. These layers included intrapersonal risk/benefit beliefs (severity, susceptibility, benefits/barriers), interpersonal social networks, institutional forces (mass media platforms, perceived uncertainty, distrust in institutions, political influences), and structural constraints (economic barriers). This framework highlights the importance of targeted approaches focusing on the dominant factors within each community to improve vaccination uptake. This concept is also supported by another study (Brumbaugh et al, 2025) reporting the best strategies to improve vaccine uptake are population-specific and context-sensitive rather than universal. This study also reported that the best interventions often integrate community engagement, culturally/linguistically appropriate messaging, trusted messengers, and strong provider recommendations, supported by better local data to guide targeted outreach. Factors including misinformation (especially on social media) and structural barriers like cost, education, transportation, and parental hesitancy (Richman 2025, Cagnotta 2025) also impact vaccine uptake, thus attitude change alone may not translate into vaccination if access barriers remain. Public health agencies typically monitor vaccination coverage at the state or regional level. However, this approach has limited guidance for undervaccinated adolescents across different socioecological populations. As resources for immunization programs remain constrained, the need for a data-driven approach that can identify those at the highest risk of being vulnerable and guide efficient outreach efforts seems necessary. Hence, it is important to develop a data-driven framework that identifies factors significant in predicting an adolescent's likelihood of staying up to date on vaccinations. Understanding such factors can help in designing targeted interventions and risk-tailored outreach strategies to empower healthcare professionals with data-driven insights and ensure high vaccination rates.

Conceptual Framework

We adapted the Social Ecological Model (SEM) as our theoretical conceptual framework to understand vaccination completion among adolescents. The SEM organizes determinants of health behavior across five domains: individual, interpersonal, organizational/provider, social/community, and policy (Bronfenbrenner 1977, McLeroy 1988). Since vaccine

completion among adolescents is a multilevel issue that involves factors beyond adolescent's individual characteristics such as parental and provider engagements, the SEM provided the comprehensive structure to understand these factors (Figure 1). The framework informed selection of demographic, socioeconomic, healthcare-access, and provider-level variables available in the NIS-Teen dataset to identify factors associated with adolescents being up to date on recommended vaccinations (Figure 1).

Machine-Learning Approach

Traditional analytic approaches, such as standard logistic regression with manual or stepwise variable selection, may be limited in their ability to model complex, multilevel relationships among demographic, socioeconomic, healthcare-access, and provider-level factors that impact adolescent vaccination completion. Advanced machine-learning methods, such as Least Absolute Shrinkage and Selection Operator (LASSO) regression and gradient-boosted decision trees, allow efficient handling of complex survey data through analyzing interactions, automated feature selection and regularization. These approaches allow identification of the most significant social-ecological predictors while reducing overfitting, making them appropriate for classifying adolescents into meaningful vaccination-risk tiers using large, nationally representative datasets such as NIS-Teen.

Research Question

What multilevel socioecological factors are associated with adolescent vaccination completion rates, and how can these factors be used to classify adolescents into meaningful vaccination-risk tiers using NIS-Teen data?

Hypothesis

Adolescent vaccination completion is associated with multilevel socioecological factors including individual, interpersonal, organizational, and community domains, and these factors can be used to classify adolescents into distinct vaccination-risk tiers that exhibit meaningful differences in population characteristics and outreach needs.

Objectives

1. Using NIS-Teen Data (2023), develop a predictive model to estimate the likelihood of adolescents staying up to date on recommended vaccinations and evaluate the model performance using training and validation datasets
2. Assign individuals to a specific risk tier using the predicted probabilities of being UTD with recommended vaccinations
3. Evaluate the accuracy of risk tiers by comparing predicted risk to actual vaccination rates to assess calibration and determine whether each tier reflects meaningful differences in the vaccination rates
4. Identify multilevel social-ecological factors that drive vaccination completion and build adolescent social-ecological profiles associated with different risk tiers
5. Develop vaccination outreach strategies by vaccination risk tiers integrating results from the machine-learning model with guidance from public-health experts to enhance real-world applicability

Methodology

Source Data and Data Acquisition

The dataset included adolescents aged 13-17 years from the most recent National Immunization Survey (NIS-Teen) data (2023). NIS-Teen is an annual survey that monitors vaccination coverage among these adolescents through two phases. First, the members of the survey team randomly call households containing eligible adolescents, and survey the parents or guardians. Then, the surveys are mailed to the adolescent's vaccination providers to get the vaccination history of the adolescent. Data includes various demographic, socioeconomic, behavioral, and provider-specific characteristics including but not limited to age, sex, race/ethnicity, insurance, poverty index, education, provider practices, geographic region, and vaccine records. These de-identified data do not contain any personal information that can identify the adolescents or their families. The data files are available to download from Centers for Disease Control (CDC) here - <https://www.cdc.gov/nis/php/datasets-teen/index.html>

This project did not involve any physical experiments, human subjects, biological materials, chemicals, or devices. All analyses were conducted using publicly available, fully de-identified secondary data from the CDC's NIS-Teen surveys, which is exempt from IRB

oversight and poses no risk to participants. The methodological workflow was entirely computational.

Estimates from 2023 NIS-Teen Data

A total of 17,021 adolescents aged 13-17 years who were born during January 2005–December 2010 and with adequate provider data were included in the analysis. Adolescents with receipt of one or more non COVID-19 vaccine doses reported by a provider and those who had not vaccinations reported by the parent or guardian were considered to have adequate provider data. The reported household response rate was 24.4%, and 39.5% of adolescents with completed interviews had adequate provider data.

Primary Outcome Variable

The primary outcome was a binary indicator of up to date vaccination for recommended adolescent vaccines excluding influenza and COVID-19. This variable was coded as 1 if the adolescent's provider confirmed up to date vaccination for the adolescent vaccine series ($P_UTD13212 > 0$) and HPV ($P_UTDHPV > 0$), and 0 otherwise. A respondent (adolescent) was considered up to date (UTD) with recommended adolescent vaccines if both of the following conditions are met, based on the provider reported vaccination data:

1. $P_UTD13212 = 1$

Indicates the teen is up to date with the 1:3:2:1:2 vaccination series, derived from provider-verified records. The 1:3:2:1:2 series refers to (ACIP 2025): 1 dose of Tdap (Tetanus, Diphtheria, Pertussis), 3 doses of Hepatitis B (HepB), 2 doses of MMR (Measles, Mumps, Rubella), 1 dose of MenACWY (Meningococcal conjugate), and 2 doses of Varicella (Var). Vaccinations administered after the Random Digit Dialing (RDD) interview date were excluded, ensuring the measure reflects only vaccinations completed before or on the interview date.

2. $P_UTDHPV = 1$

Indicates completion of the HPV vaccine series per ACIP recommendations, based on age and dose timing (2-dose if initiated before age 15; 3-dose if initiated at 15 or older).

A respondent was coded up to date with the recommended vaccinations (UTD = 1) if both criteria above are satisfied. The COVID-19 vaccine and influenza (flu) vaccine were excluded from the UTD definition. This is because they have different public health objectives, as COVID-19 and flu vaccines targeting seasonal or pandemic specific protection, whereas the 1:3:2:1:2 + HPV series represents the core adolescent immunization schedule recommended by the CDC/ACIP for routine preventive care (ACIP 2025). Any records with missing data on the primary outcome variable were excluded.

Predictors

Predictor variables included demographic factors (age, sex, race, ethnicity, language), socioeconomic factors (income, home ownership, education), measures of behavior and access (well-child visits, number of visits, prior vaccination behaviors, access to providers, type of facility), regional factors (state, census region, mobility), and measures of provider engagement (HPV recommendation, checkup at 11-12 years). These variables were selected based on the theoretical framework, prior research and expert guidance.

For modeling purposes, we converted categorical variables to indicator variables via one-hot encoding by producing a separate binary variable for each category level. This encoding allowed inclusion of multilevel survey variables (state, facility type, caregiver type, income bands, etc.) within a single linear model. Missingness in select variables was retained by creating explicit “missing/unknown” levels allowing the model to learn whether absence of information was systematically associated with the study outcome.

Statistical Analysis

The entire coding was done in Google Colab (Python 3.12), starting with data preparation and variable construction. For modeling purposes, the data were split into training (70%) and validation (30%) subsets using stratified random sampling to maintain proportional representation of vaccination status. We followed the approach described in a previous study (Skyles 2024) to identify the optimal predictive modeling approach for our study. Specifically, we used the LazyClassifier function from the lazypredict (Shankar Rao Pandela, Bengaluru, India v0.2.10) package to train and evaluate multiple machine-learning models from the Scikit-learn (1.4.1) package with default parameters. Models were compared on a validation set

using performance metrics including accuracy, balanced accuracy, ROC-AUC, and F1 score. Logistic-family models performed the best (Figure 2). Next, the hyperparameter tuning was performed on both the logistic regression models (L1 and L2-regularized) using the Optuna (3.6.0) package to maximize mean five-fold cross-validated AUROC. Tuning did not improve AUROC compared with default regularization parameters; therefore, the final model retained the cross-validated optimal configuration. Based on the results, we selected the L1-regularized LASSO (least absolute shrinkage and selection operator) regression model because it provided comparable predictive performance while producing a sparse set of predictors, facilitating automated variable selection and alignment with the SEM Model framework. The performance of the final selected model was evaluated using confusion matrices, sensitivity, specificity, precision, and recall.

To identify the most influential predictors, we extracted feature names and coefficients from the final LASSO model. Nonzero coefficients were interpreted as predictors retained by LASSO regularization, and predictors were ranked by absolute coefficient magnitude. Coefficient sign indicated direction of association with being UTD (positive = higher odds; negative = lower odds), conditional on other included variables. These significant predictors were mapped to the domains of the SEM model (structural, community/geography, organizational, interpersonal/family, and individual) to provide a core understanding of their contributions.

Finally, using the distributions of predicted probabilities of an adolescent being UTD with the recommended vaccinations, we constructed population level risk tiers. The performance of these tiers was analyzed by comparing actual vs predicted outcomes. Each risk tier was characterized using the significant characteristics informed by the SEM and machine-learning models to make those tiers actionable in real-world settings.

Code and Data Availability

The final code used to process the NIS-Teen dataset, select and train a model, identify important features overall, and by risk tiers is available on Github.

Results

Predictive Model Performance

A total of 17,021 adolescents were included in the analysis. On the validation dataset (n=5,107), the model achieved an AUROC of 0.68 with a balanced accuracy of 0.63. Using a fixed classification threshold, the model yielded an F1 score of 0.73, driven by relatively high precision (0.79) and moderate recall (sensitivity = 0.67). Model specificity was 0.58, indicating a moderate ability to correctly identify adolescents who were not UTD. The confusion matrix reported that among adolescents predicted to be UTD, 2,421 were true positives and 635 were false positives, while 1,171 false negatives and 880 true negatives were observed.

SEM-aligned Significant Predictors

A total of 33 unique predictors were retained by the model (101 non-zero categorical features through one-hot encoding method). The influential variables are displayed in Figure 4. We aligned top predictors identified by the final model across multiple domains of the SEM framework. The majority of these predictors belonged to the organizational and structural domains. Factors such as healthcare access and delivery, particularly indicators of preventive care engagement emerged as the strongest positive predictors of an adolescent being UTD with recommended vaccinations. Adolescents who had completed a well-child visit, received an 11–12-year preventive checkup, and had a documented provider recommendation for HPV vaccination were associated with higher UTD rates. The strongest negative predictors included missing or non-traditional facilities, lack of recent healthcare visits, absence of an early adolescent preventive checkup, and lack of provider recommendation were strongly associated with lower predicted probability of being UTD. Geographical regions were impactful with several state and regional indicators showing strong associations with vaccination status. Residence in certain states and in southern or western census regions was associated with lower predicted likelihood of being UTD, while other states showed positive correlations. Finally, individual and family-level factors also played a crucial role. Household structure, language spoken in the home, residential mobility, and socioeconomic indicators such as poverty status and parental education were associated with vaccination status. For example, higher residential mobility and single-child households were associated with lower predicted UTD probability,

while Spanish language households and multiracial identity were associated with higher predicted probability of UTD.

Risk Tiers and Outreach Strategies

Based on the empirical distributions, predicted probabilities of an adolescent being UTD with the recommended vaccinations were categorized into five risk tiers: Very Low ($\geq 80\%$ predicted probability), Low (70-79.9% predicted probability), Moderate (60-69.9% predicted probability), High (50-59.9% predicted probability), Very High ($< 50\%$ predicted probability). Thus, lower risk tiers indicated higher probability, while higher risk tiers reflect a lower probability of being UTD with vaccinations. The performance of five risk tiers was analyzed by comparing the observed UTD rates among each risk tier (Figure 5).

The characteristics and recommended outreach strategies by risk tiers are listed in Table 1. Adolescents in the very low risk tier indicating the highest predicted likelihood of being UTD showed the greatest vaccination uptake, with 85.5% up to date. These adolescents were more likely to maintain consistent engagement with preventive care, including routine well-child visits, and multiple touchpoints in clinical settings. This risk tier can be adequately addressed with routine reminders and reinforcement through existing health care and school-based systems. The low risk tier group also had a relatively high observed vaccination rate (79.4%), however showed a greater variability in health care engagement. This group was primarily characterized by less frequent missed well-child or early adolescent preventive visits, and predominantly received care in private clinical settings with inconsistent provider engagement. For this group, outreach strategies that may prove beneficial include low-intensity reminder and follow-up mechanisms integrated into routine pediatric and school health workflows. In the moderate risk tier group, observed UTD rates declined further (68.1%), and adolescents were more likely to have mixed racial and ethnic profiles, inconsistent attendance at preventive care visits, disrupted continuity (e.g., moving from a different state and non-biological/female guardian caregiving), and regional/geographic variation.. This group represents a population with persistent missed opportunities for vaccination for whom consistent and targeted reminders, school or community engagement, and strategic provider messaging may yield substantial improvements in vaccination uptake. Adolescents classified in the high risk tier experienced substantially lower vaccination coverage (48.0%) and greater structural barriers to care. These adolescents were

primarily from lower household income, higher residential mobility, and with larger gaps in documented regular visits and checkups. This group faced continuity challenges such as higher residential mobility and caregiver transitions (e.g., step/foster/adoptive/female guardians), and more frequent care in non-traditional settings (school/teen clinics or other facilities). The need for this tier can be addressed through more formal outreach strategies such as personalized bilingual outreach, support for appointment scheduling, help with transportation and alerts for providers. Finally, the very high risk tier adolescents had the lowest observed UTD rates (27.3%) and exhibited the most significant structural and systemic barriers to vaccination. This group was characterized by higher prevalence of below-poverty households, high residential instability, limited contact with health care providers, and minimal engagement with traditional preventive care systems. The tier was characterized by major missed opportunities within the healthcare system, including lack of documented HPV vaccine recommendation and missing or non-traditional facility patterns. This group potentially requires the most intensive outreach strategies with consistent engagement of community health workers, school/church vaccine drives, home visits and wraparound support, and culturally tailored messaging.

Discussion

The data-driven frameworks integrating insights from the machine-learning models with the elements of socio-ecological (SEM) model can provide actionable insights for the vaccination outreach. Although its discrimination (AUC 0.68) indicates moderate predictive accuracy, it showed a stronger ability to correctly identify adolescents who were UTD with recommended vaccinations than those who were not. Overall, the results from the predictive model indicated strong ability to capture vaccinated individuals and moderate false-positive rate. Thus, the model prioritized correct identification of UTD adolescents at the expense of some misclassification among non-UTD individuals, consistent with its intended use as a population-level risk stratification tool rather than a diagnostic classifier.

Through the feature engineering process, the model also identified a set of stable predictors impacting the vaccination behaviors among adolescents. Using the predicted probabilities from the model, we stratified populations into five actionable tiers that can help in designing targeted outreach strategies. The decreasing UTD rates across increasing risk tiers validates the model's discriminative performance and highlights its usefulness in identifying

at-risk adolescents with the highest need of targeted outreach and/or healthcare provider follow ups. These tiers can inform resource allocation and improve cost-effectiveness of vaccination campaigns.

Model Performance in Context of Existing Literature

The performance of our model is consistent with findings in the current literature that used a predictive modeling approach to assess vaccination behaviors among various populations. Prior studies utilizing logistic regression or machine-learning techniques to predict vaccination rates among adolescents and adult populations reported AUC values ranging from 0.65 to 0.82, reflecting the inherent difficulty of predicting vaccination behavior from observational data. For example, a recent study by Petrik et al. (2024) developed an HPV vaccination completion risk model using demographic and clinical characteristics with an AUC of 0.65 among children aged 11-17 years in a large integrated delivery system in Oregon (Petrik et al, 2024). Oster et al. (2021) developed a newborn undervaccination risk model using electronic medical records linked with Washington State immunization registry data and reported a composite AUC of 0.67 across five risk prediction models. Cheong et al. (2021) used machine-learning methods including a hypertuned XGBoost regressor to predict county-level COVID-19 vaccination uptake and reported a 62% cross-validated accuracy, demonstrating the typical level of predictive performance achievable using large-scale survey or ecological data. Most relevant to our study, Mathkor et al. (2025) applied advanced machine-learning and AI methods including Balanced Random Forest, XGBoost, and SHAP to a sample of 27,565 NIS-Teen records. Although this study was not aimed at building predictive models, their results highlighted geographic location, vaccine type, poverty level, and race/ethnicity as the strongest predictors of adolescent vaccination coverage, mirroring several of the key predictors identified in our model, such as state of residence, poverty, mobility, and provider engagement. The hybrid machine learning and rules-based model from Kierner et al. (2025) achieved AUC values ranging from 0.59 to 0.69, with the best-performing ensemble method yielding the AUC of 0.69, nearly identical to the AUC of 0.68 achieved in our model (Kierner 2025). This further supports that vaccination behavior prediction, even when combining clinical rules with machine learning, typically achieves moderate discrimination consistent with the performance of our NIS-Teen model. A few studies conducted outside the U.S reported relatively higher AUCs. For example, a large

feasibility study using 47,554 longitudinal immunization records from low- and middle-income countries in Singh province reported substantially higher discrimination when richer clinical and visit-timing data were available. In that study, recursive partitioning achieved an AUC of 0.79 (95% CI: 0.78–0.80), support vector machines reached 0.79, conditional forests reached 0.78, and random forests achieved 0.75 (Chandir 2018). These results illustrate that models built on detailed longitudinal clinical data can outperform models based on cross-sectional survey data like NIS–Teen, which inherently capture fewer behavioral and system-level nuances. Another study predicting measles vaccination dropout in Ethiopia using Demographic and Health Survey data achieved an XGBoost AUC of 0.81 (with Random Forest AUC = 0.82 and Support Vector Machine AUC = 0.64) (Alemayehu 2024). These higher AUCs are consistent with settings where immunization timing, visit histories, and clinical records are directly captured. Within this context, our model’s AUC of 0.68 using cross-sectional survey data falls well within the expected performance range and is comparable to, or slightly better than, many published baseline models. Most importantly, despite moderate AUC values, our predictive model achieved high recall and strong risk stratification, enabling reliable identification of adolescents with systematically low vaccination uptake. Overall, our results align with broader evidence supporting machine-learning tiering systems as effective tools for triaging and prioritizing at-risk adolescents falling behind on recommended immunizations.

Socioecological Determinants of Adolescents Vaccination Completion Status

Overall, the final model’s predictors demonstrate a clear multilevel structure consistent with the SEM framework. These findings align with the previous studies showing the impact multilevel determinants on vaccine uptake rates (Li 20205, Vardavas 2023, Sacre 2023, Sonawane 2024, Enyinnaya 2024, Richman 2025, Cagnotta 2025, Brumbaugh 2025). In our study, while individual and family characteristics contributed to vaccination risk, the most important predictors were related to healthcare system engagement, provider recommendation, and access to preventive services, reinforcing the importance of system-level interventions to improve adolescent vaccination coverage. The predictors selected by the LASSO regression model aligned with our theoretical framework adapted from the SEM. The model identified various individual, interpersonal, provider, social/community level, and structural factors associated with adolescent vaccination UTD status in NIS-Teen data. The SEM additionally

includes policy and systems level influences, which informs vaccination access and delivery through national immunization guidelines and public health infrastructure. Although these factors were not directly measured in the analytic dataset, their effects are assumed to operate uniformly across the study population.

An empirical goal of our study was to predict adolescent UTD vaccination status from socioecological variables and translate model outputs into risk-tier profiles that can guide feasible, real-world public health outreach. Stratifying adolescents by predicted probability revealed a clear pattern of barriers limiting UTD status. Very low and low risk tiers reflect a routine-care connected profile, characterized by stable family context and consistent preventive care engagement; for these adolescents, maintaining high coverage is likely achievable through reinforcement of routine practices and low-burden reminder systems. Moderate risk tiers represent a profile that highlights missed opportunity, with more heterogeneous family and demographic contexts and less consistent preventive care participation suggesting that undervaccination often arises from fragmented followthrough rather than complete disengagement, making communication and reminders during provider visits valuable. Across tiers, organizational and provider factors meaningfully differentiated outcomes. When preventive visit cadence is inconsistent or care shifts toward public/community-based settings, reliable recommendation and completion opportunities become less predictable, highlighting provider access and visit frequency as actionable points for intervention. In high and very high risk tiers, barriers are increasingly structural and system-level, aligning with a care discontinuous profile impacted by mobility, housing and economic instability, and limited integration into traditional preventive care. Improving UTD vaccination in these groups likely requires outreach that combines logistical support (navigation, scheduling facilitation, transportation/coordination) with culturally appropriate engagement delivered through community-based settings. Finally, for the highest risk populations, characterized by minimal access and contact with traditional health care systems, outreach strategies extend beyond clinical settings and rely on community based approaches addressing care gaps.

Limitations

This study has limitations inherent to retrospective database studies. NIS-Teen household response rates were low, and only 39.5% of the households interviewed had adequate provider

data, potentially indicating selection bias. Second, residual bias may remain due to incomplete determination of vaccination status. Next, total survey error assessments indicate that NIS-Teen may underestimate true vaccination coverage, particularly for HPV, which could affect estimates of UTD status. While NIS-Teen utilizes complex sampling weights for population-level prevalence estimates, we excluded these weights since our objective was to identify individual-level risk patterns rather than population-level estimates. Nonetheless, this may limit the generalizability of the derived risk to the broader U.S. adolescent population, hence future work should evaluate weighted or survey-adjusted modeling approaches to assess generalizability. Finally, several predictors used in the model such as provider recommendation documentation, facility type indicators, and mobility or socioeconomic variables may not always be available in real time within routine clinical workflows. Thus, implementing the outreach tool in practice may require further integration with electronic health record systems or public health registries where these variables might be reported.

Table and Figures

Figure 1: The Social Ecological Model for Understanding Vaccine Behavior

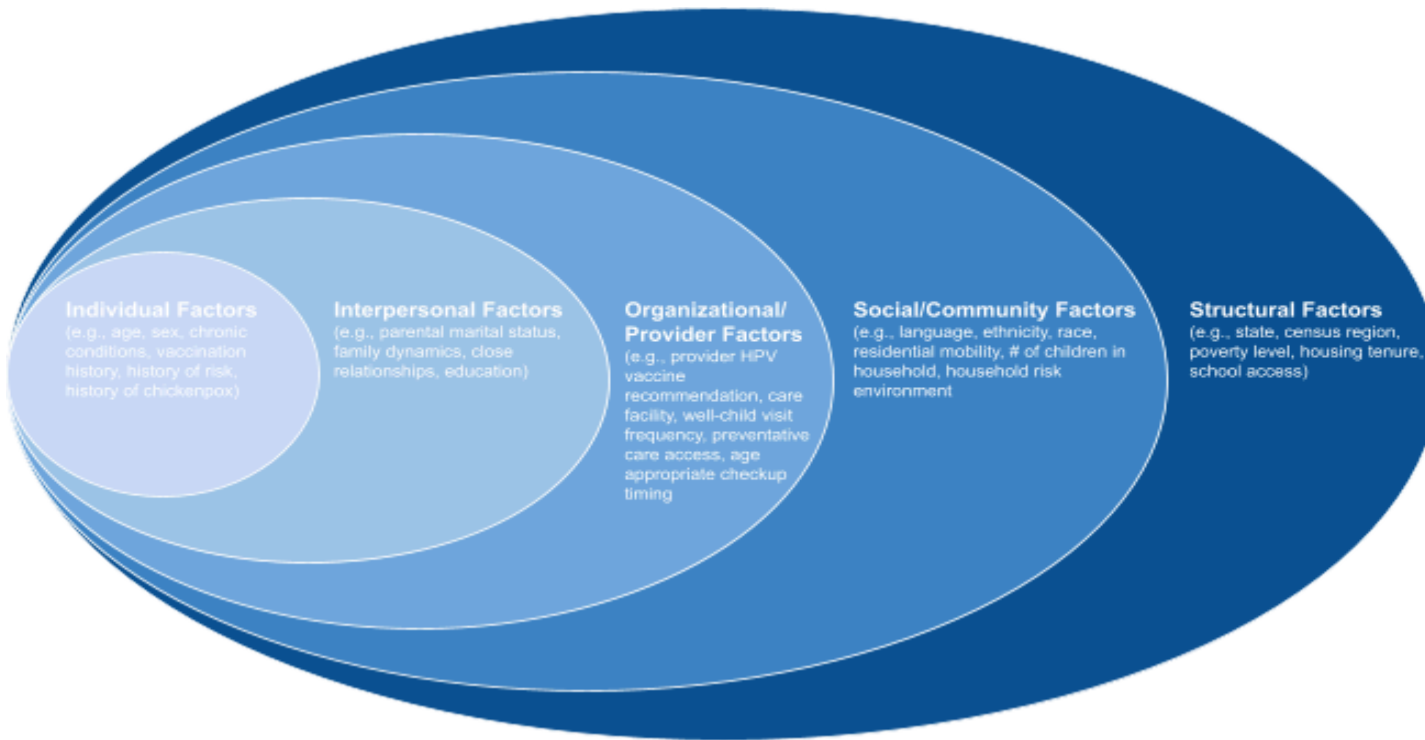


Figure 2: Performance of Candidate Predictive Models

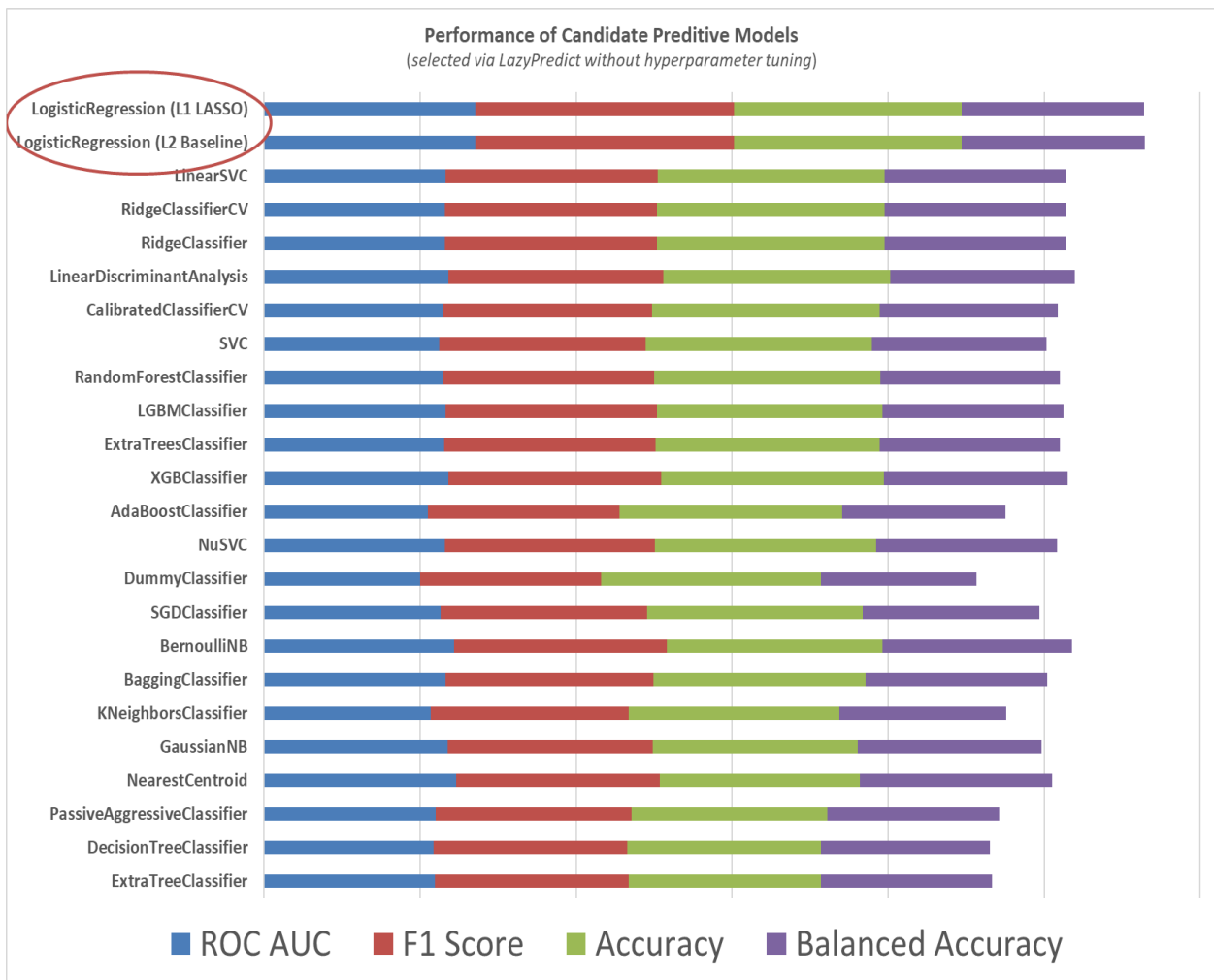


Figure 3: The Final Selected LASSO-Regression Model Performance

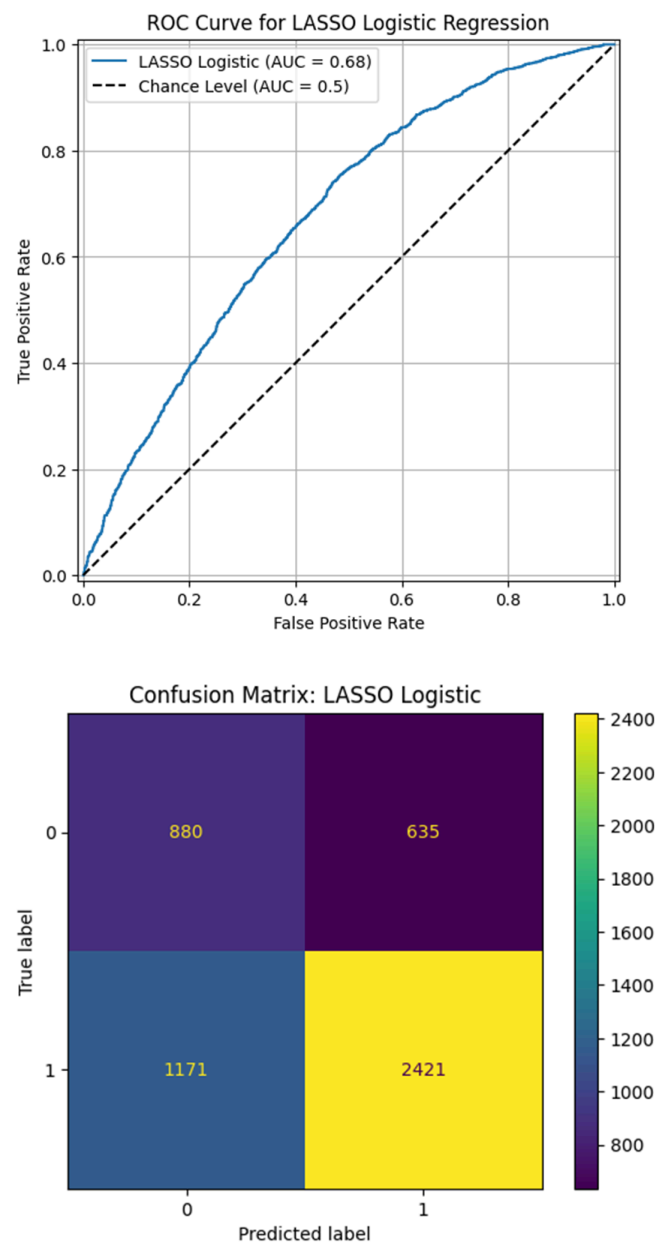
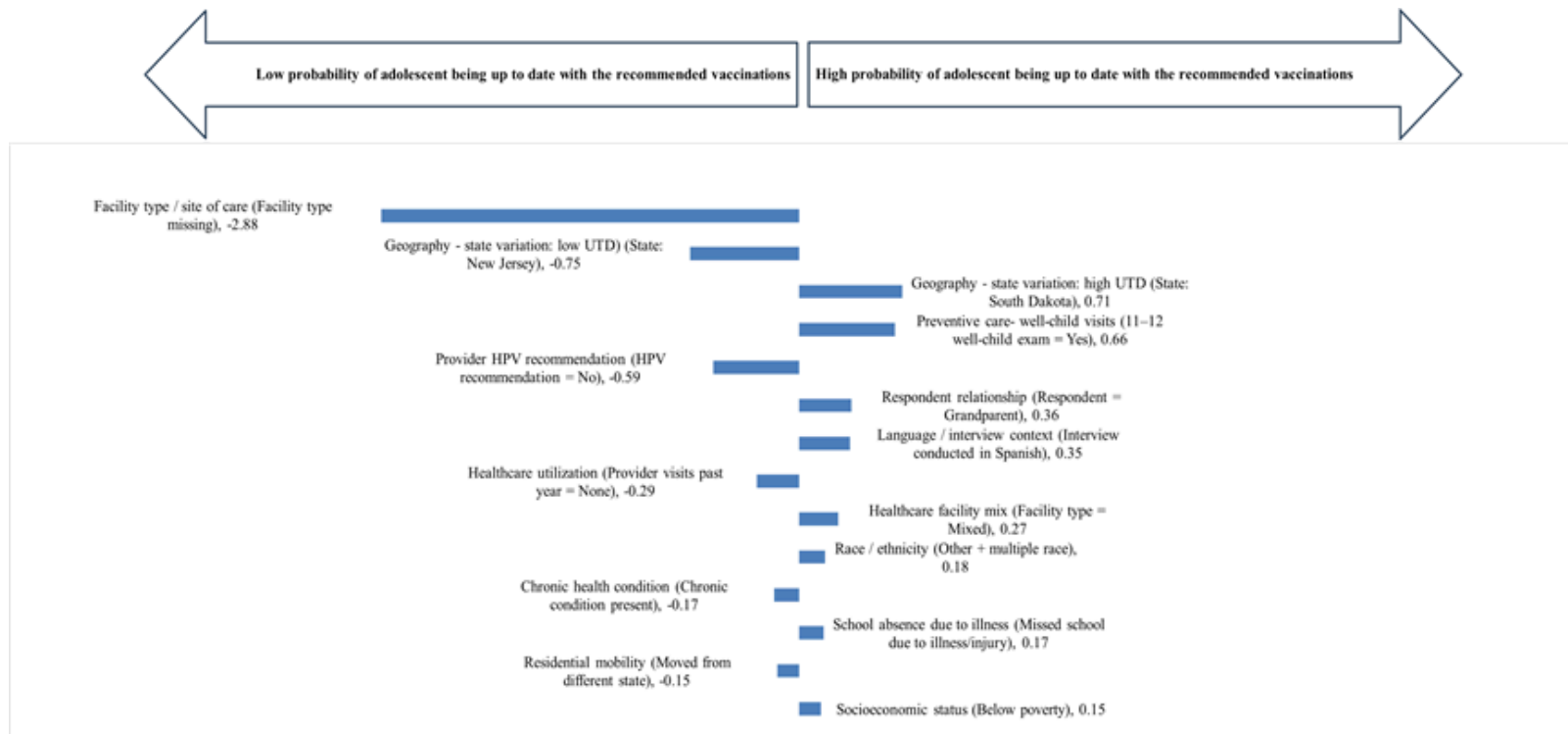


Figure 4: Predictors from the L1-regularized logistic regression model



Related one-hot encoded feature levels were grouped into broader predictor themes, and the largest absolute coefficient within each theme was used as a representative coefficient. Representative individual feature levels with their signed coefficients (log-odds scale). Positive coefficients indicate association with higher predicted probability of being up to date with recommended adolescent vaccinations, while negative coefficients indicate lower predicted probability.

Figure 5: Risk Tiers: Actual vs Predicted:

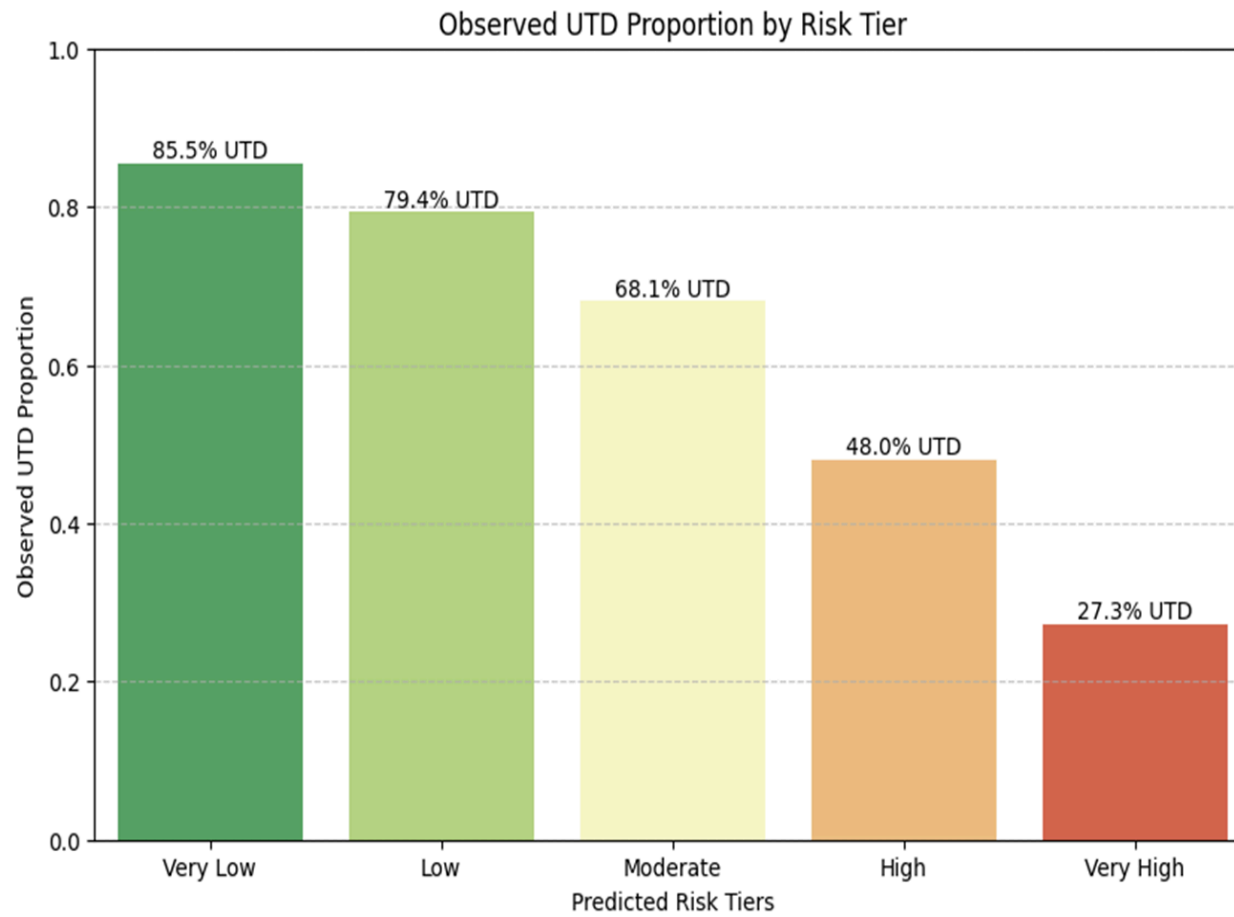


Table 1: Vaccination Outreach Risk Tiers, Characteristics, and Strategy

Risk Tier	Key Socio-Ecological Characteristics	Potential Outreach Strategy
Very Low Risk	Consistent preventive-care touchpoints (well-child + 11–12 checkup); documented HPV recommendation; care across stable/consistent settings; few continuity disruptions.	• Annual reminders (text/email) • Booster updates • Reinforcement via schools/providers
Low Risk	Mostly engaged with preventive care, but early missed-opportunity signals emerge (occasional gaps in visits/checkups or weaker recommendation documentation); some regional effects (South/West).	• Light touch reminders • Pediatrician/school nurse follow-up • Provider prompts: “Recommend every time”
Moderate	Mixed provider engagement: some preventive care contact but inconsistent visit documentation; HPV recommendation gaps appear; mobility and caregiver complexity more common	• Targeted reminders (text + call) • School/community campaigns • Provider scripts reinforcing HPV & flu recs
High	Strong missed opportunity patterns with no recorded visits and/or missed annual checkups; HPV recommendation often absent; more care in non-traditional/community settings (e.g., school/teen clinics/other); mobility and caregiver transitions more common; South/West signals.	• Personalized bilingual outreach • Appointment scheduling support • Transportation help
Very High	Minimal contact with preventive care; highest levels of missing documentation (e.g., facility missing); HPV recommendation absent; mobility (including cross-state) + Southern geography prominent; multiple systemic barriers.	• Community health workers • School/church vaccine drives • Home visits and wraparound support • Culturally tailored messaging

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Code Availability

The code used for data preprocessing, feature engineering, model training, and evaluation is publicly available in a GitHub repository for transparency and reproducibility.

Repository: [https://github.com/\[repository-name\]](https://github.com/[repository-name])

A **QR code** linking to the repository is provided below for quick access.